

Mechanisms of Obesity and Obesity Management

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Disclosures

I have no financial disclosures nor potential conflicts of interest to report.

Learning Objectives

1. Explain the regulation and pathophysiology of appetite and energy balance that contribute to obesity
2. Formulate a therapeutic management plan to treat obesity
3. Describe the clinical use, contraindications, mechanism of action, side effects and efficacy of each of the FDA-approved medication options used to treat obesity
4. Identify which patients may benefit from surgical options to treat obesity

Outline

1. Define obesity as a serious, chronic, progressive disease
2. Discuss the physiology of hunger as we understand it today
3. Discuss diagnosis and evaluation
4. Discuss treatment options
 - a. Behavioral
 - b. Pharmacologic
 - c. Surgical

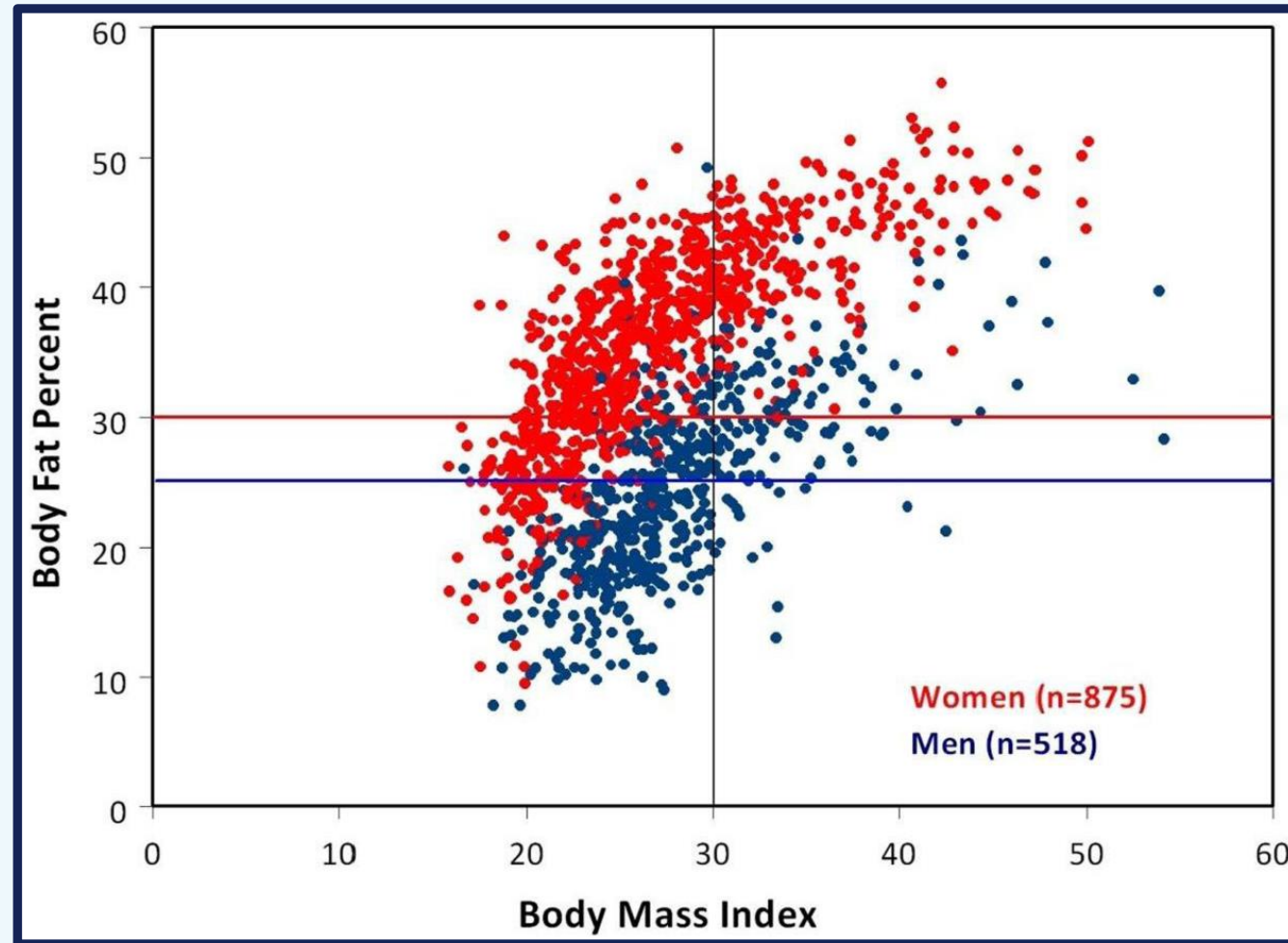


Definition of Obesity

Obesity is a chronic disease characterized by excess adipose tissue deposition caused by a disruption of the normal physiology of energy balance resulting in many potential complications, many of them debilitating and/or life-threatening.



Body Mass Index as an Indicator of Body Fat and Health - Limitations



Shah NR, Braverman ER (2012) Measuring Adiposity in Patients: The Utility of Body Mass Index (BMI), Percent Body Fat, and Leptin. PLoS ONE 7(4): e33308. doi:10.1371/journal.pone.0033308

Classification of Obesity - Adults

- Obesity is typically classified using body mass index (BMI)
 - $\text{BMI} = \text{total body weight (kg)} / \{\text{height (m)}\}^2$

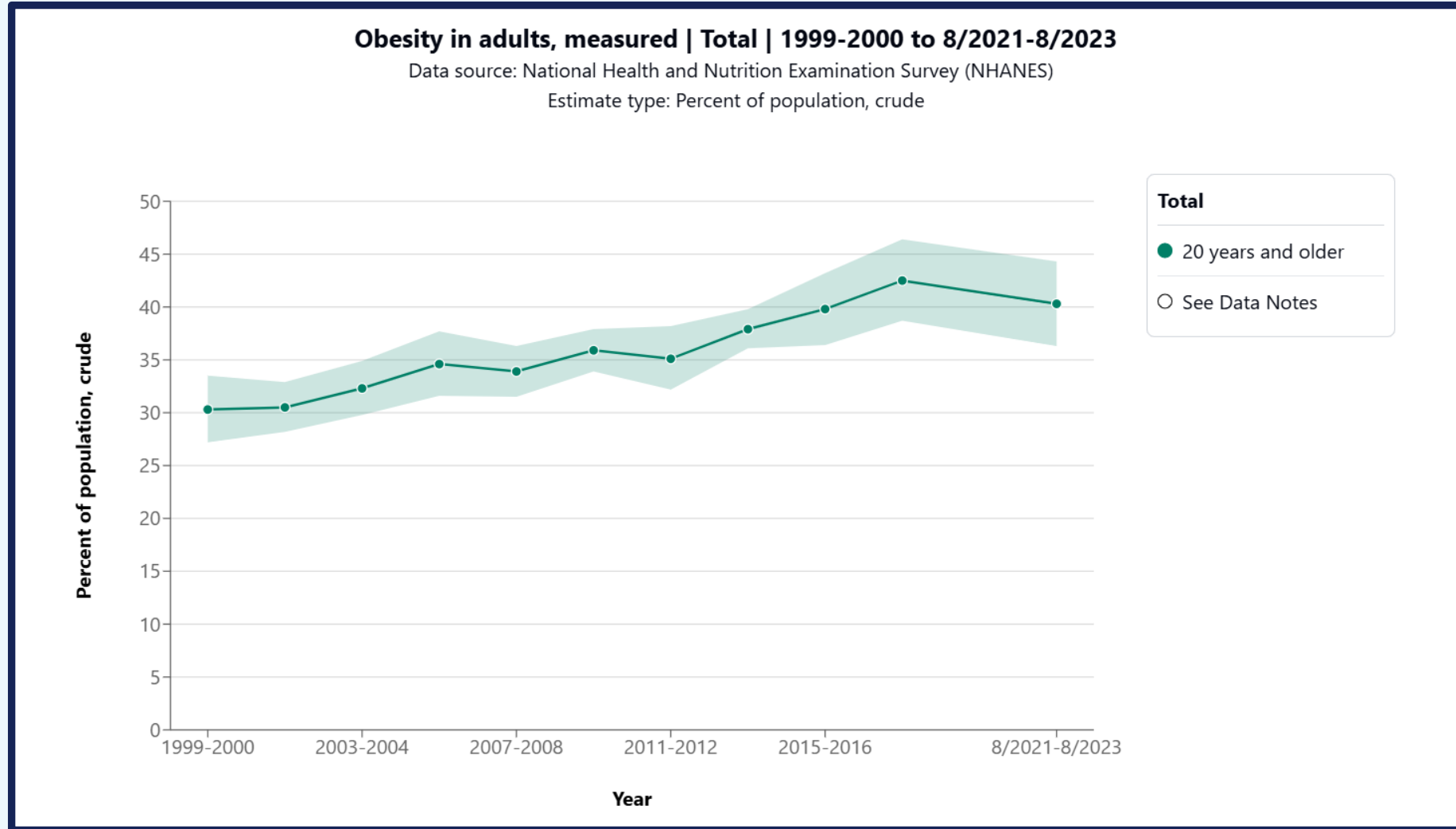
Classification	International Classification (kg/m ²)	People of Asian Descent (kg/m ²)
Underweight	≤ 18.4	≤ 17.9
Healthy Weight	18.5 to 24.9	18 to 22.9
Overweight	25 to 29.9	23 to 24.9
Obesity, Class 1	30 to 34.9	≥ 25
Obesity, Class 2	35 to 39.9	
Obesity, Class 3	≥ 40	

Classification of Obesity - Children

Classification	Criterion (BMI percentile for age)
Underweight	<5 th
Healthy Weight	5 th to 84.9 th
Overweight	85 th to 94.9 th
Obesity	≥95 th

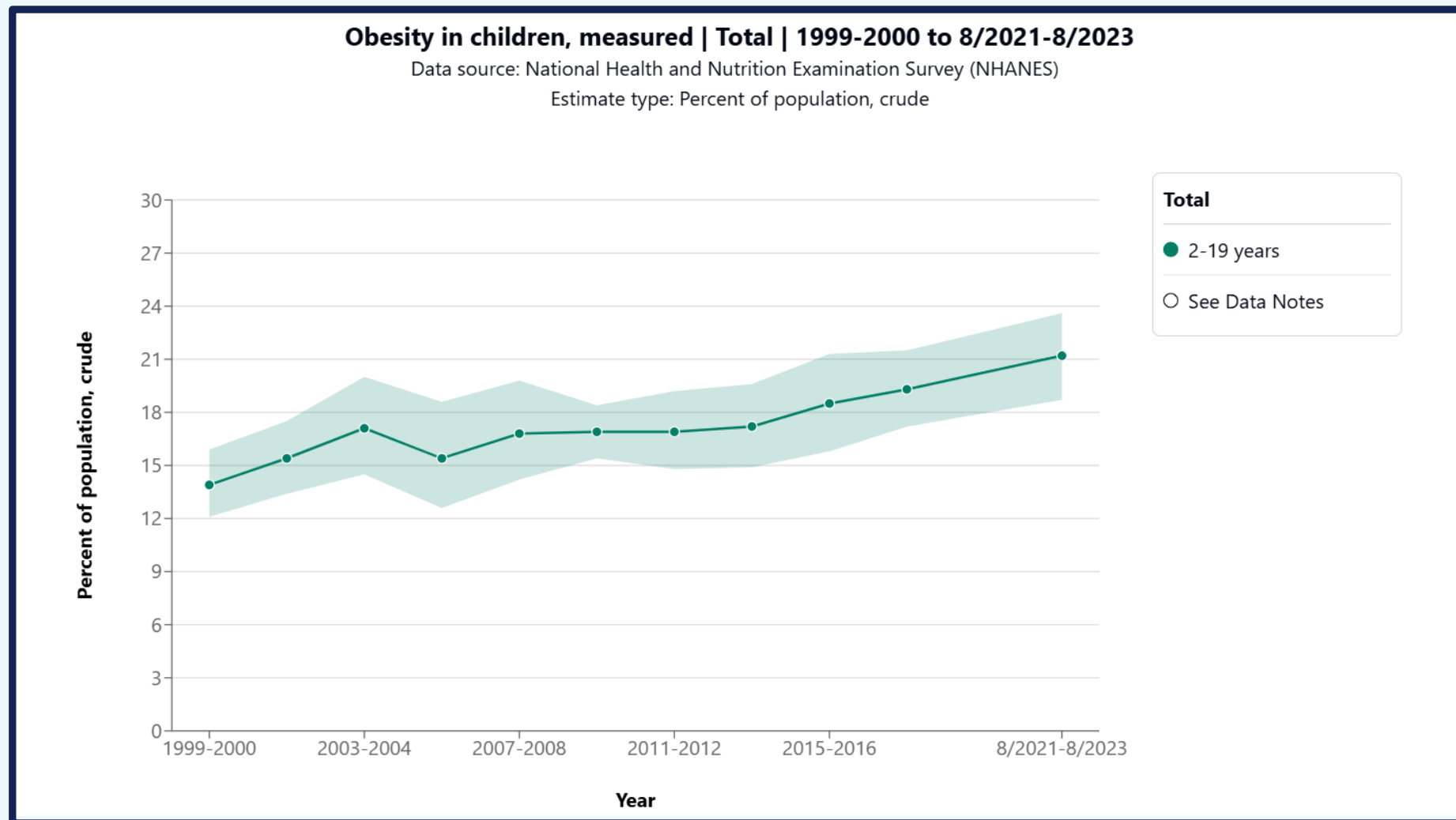


Obesity Prevalence in the US - Adults



NCHS Data Query System. Obesity in adults, measured (Internet). Hyattsville (MD): National Center for Health Statistics c2025 [cited 9/10/26]. Available from <https://www.cdc.gov/nchs/dqs/index.html>

Obesity Prevalence in the US - Children



NCHS Data Query System. Obesity in adults, measured (Internet). Hyattsville (MD): National Center for Health Statistics c2025 [cited 9/10/26]. Available from <https://www.cdc.gov/nchs/dqs/index.html>

Throwing Out Old, Inaccurate Ideas

Traditionally, many thought human body weight is determined by this equation:

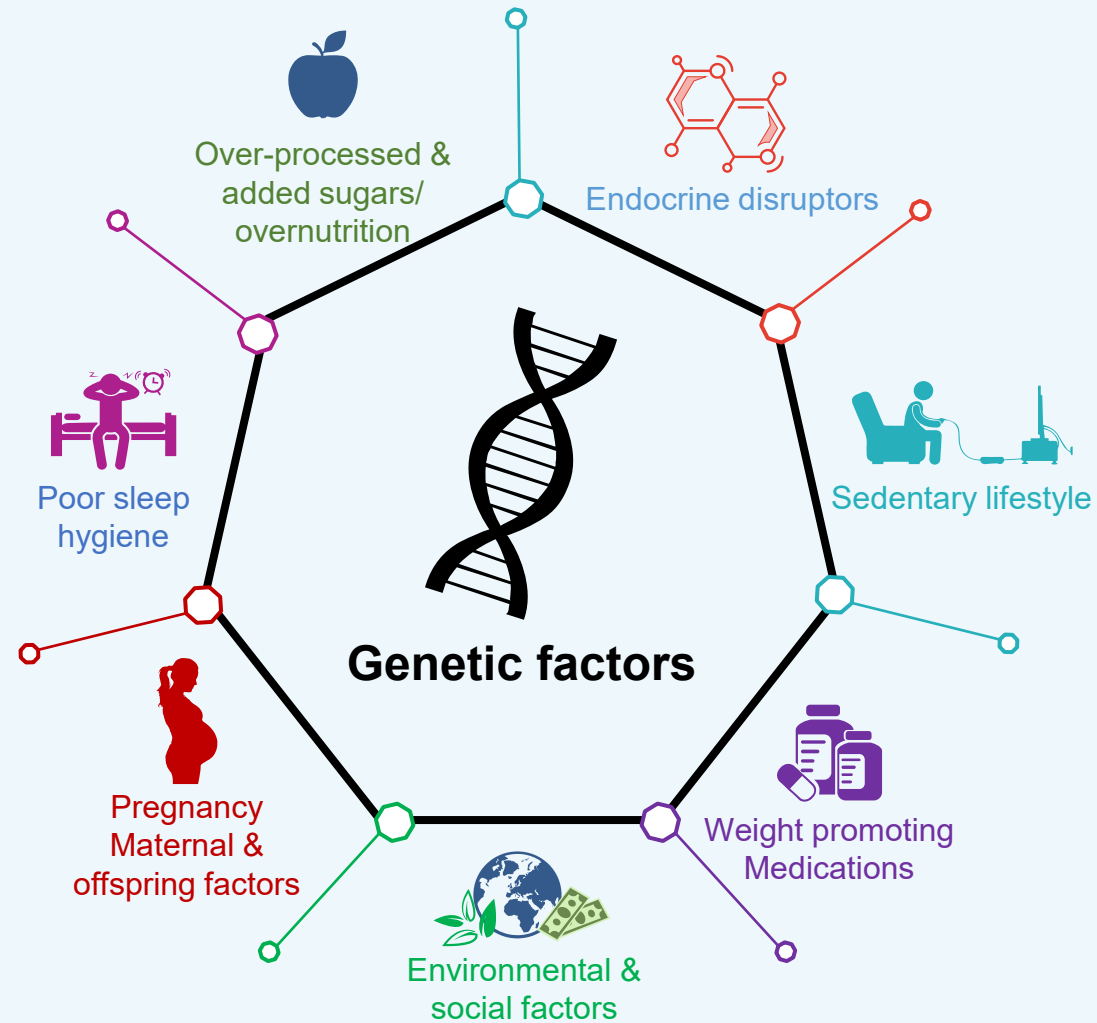
$$\Delta \text{ Body Weight} \approx \text{Calories in} - \text{Calories out}$$

We Must Set The Record Straight

Traditionally, it was thought that human body weight is determined by this simple idea:

$\Delta \text{ Body Weight} = \text{Calories in} - \text{Calories out}$

Body Weight Regulation: Our Current Understanding



Physiology of Hunger – Evolutionary Adaptation

- Many of our evolutionary predecessors were hunters and gatherers.
 - Food scarcity was the norm; starvation was a primary threat to survival.
 - Animals adapted to promote overconsumption of calories using adipose deposition as energy storage.
 - These adaptations increase survival to reproduction but have negative medical consequences.
- About 12,000 years ago, agricultural societies emerged.
 - This resulted in a significant increase in the availability of affordable, energy-dense and palatable foods.
- We harbor biology favoring caloric overconsumption and live in an environment favoring the same.



Central Biochemical Pathways – Arcuate Nucleus of the Hypothalamus

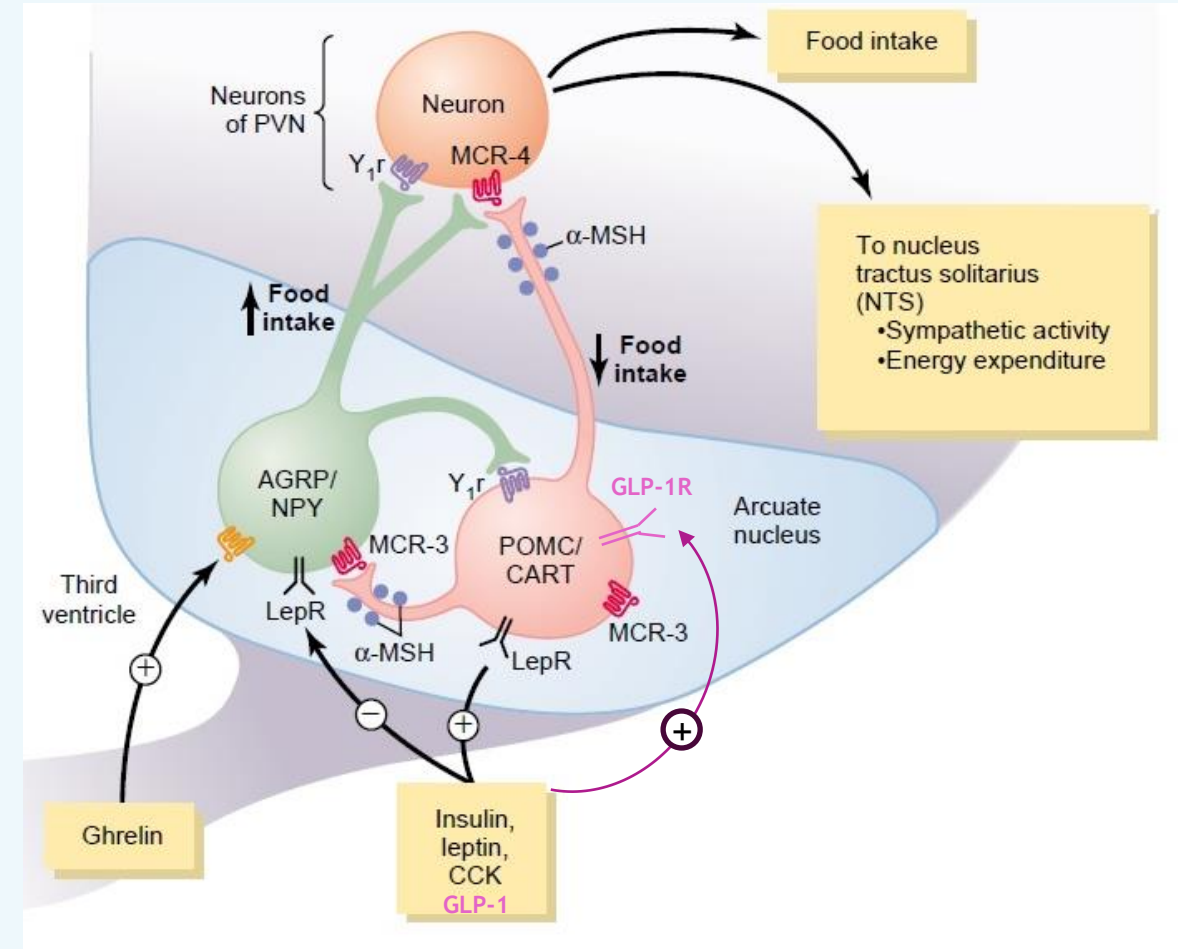
Orexigenic Pathway (left side)

- Stimulation of AGRP/NPY neurons by ghrelin leads to stimulation of the Y₁R and MC4R (receptors) resulting in increased feeding behavior and decreased energy expenditure

Anorexigenic Pathway (right side)

- Stimulation of POMC/CART neurons leads to stimulation of the MC3R & MC4R (receptors) resulting in decreased feeding behavior and increased energy expenditure

*Please resist the urge to memorize the details; they are not important for most. Please focus on the larger concepts and know there has been good work in elucidating these mechanisms.



α-MSH=α-melanocyte-stimulating hormone, CART=cocaine- and amphetamine-regulated transcript, MCR=melanocortin receptor, NPY=neuropeptide Y, POMC=proopiomelanocortin, Y₁R=neuropeptide Y1 receptor

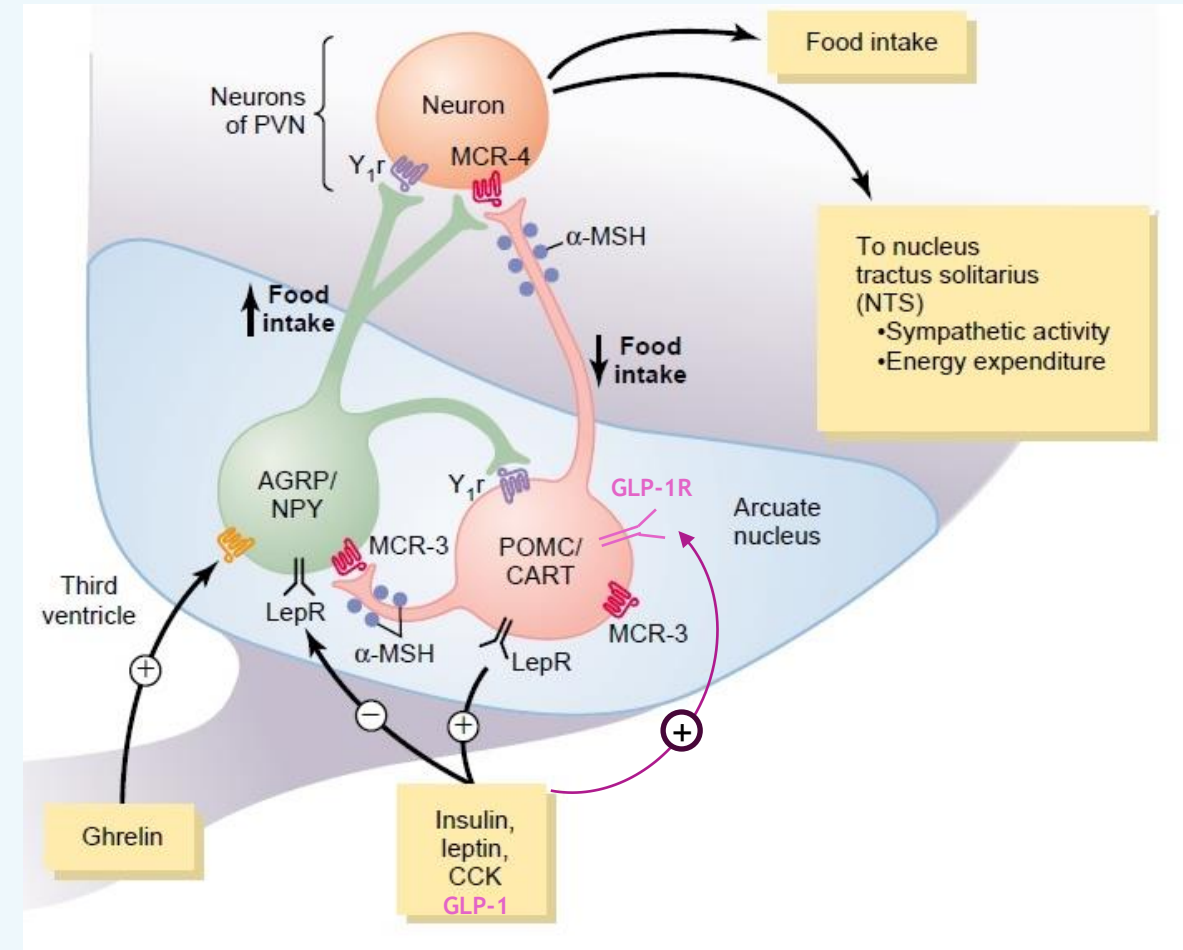
Central Biochemical Pathways – Arcuate Nucleus of the Hypothalamus

Hunger Hormones

- **Ghrelin**

Satiety Hormones (some pictured)

- Amylin
- CCK (cholecystokinin)
- **GIP** (gastric inhibitory polypeptide)
- **GLP-1** (glucagon-like peptide 1)
- **Insulin**
- **Leptin**
- OXM (oxyntomodulin)
- PYY (peptide YY)



α-MSH=α-melanocyte-stimulating hormone, CART=cocaine- and amphetamine-regulated transcript, MCR=melanocortin receptor, NPY=neuropeptide Y, POMC=proopiomelanocortin, Y₁R=neuropeptide Y1 receptor

Physiology of Hunger – Current Understanding

- 3 primary mechanisms of hunger have been identified
 - Homeostatic hunger - dominant when there is acute caloric need (in the food deprivation state)
 - Hedonic (pleasure-driven) hunger - dominant mechanism in the absence of food deprivation
 - Microbiota-driven hunger - not well-understood, likely important
- Appetite and its clinical manifestation, calorie intake, are determined by complex interactions between these mechanisms.

The Physiology of Hunger. Lee CD, Ingelfinger JR, eds. *The New England journal of medicine*. 2025;392(4):372-381. doi:10.1056/NEJMra2402679



Physiology of Hunger – Homeostatic Hunger

- Key Points
 - Dominant in the setting of acute caloric need
 - Primary mechanism is ghrelin secretion by the pancreas, small intestine and stomach
 - Low serum concentrations of amino acids, fatty acids or glucose triggers ghrelin secretion
 - Increased serum ghrelin results in hunger, increased appetite and decreased satiety via direct stimulation of the orexigenic neuronal pathway
 - Secondary mechanism is gastric mechanoreceptor quiescence which causes vagal-mediated orexigenic neuronal pathway activity
 - Generally, this does not result in excessive calorie intake

The Physiology of Hunger. Lee CD, Ingelfinger JR, eds. *The New England journal of medicine*. 2025;392(4):372-381. doi:10.1056/NEJMra2402679



Physiology of Hunger – Hedonic Hunger

- Key Points
 - “De-coupling” of appetite and acute caloric need
 - Supersedes homeostatic hunger resulting in caloric intake for pleasure when food is abundant
 - Studies suggest preferential intake of energy-dense, high-fat, and high-sugar foods when hedonic hunger is in effect
 - This type of hunger is multifactorial and extremely complex involving many poorly understood mechanisms such as learning, cognition and memory
 - Modification of hedonic hunger requires individualized strategies

The Physiology of Hunger. Lee CD, Ingelfinger JR, eds. *The New England journal of medicine*. 2025;392(4):372-381. doi:10.1056/NEJMra2402679



Physiology of Hunger – Hedonic Hunger

- Key Points

- Hedonic hunger may be triggered by many stimuli

- Negative and positive emotions cause patients to seek psychologic reward triggered by energy intake
 - Anticipation of highly-palatable food consumption (sensory inputs including food advertisements) is a powerful trigger
 - Taste preferences, cultural beliefs, and religious practices are powerful triggers
 - In some communities, obesity is considered a sign of prosperity
 - Ideas surrounding physical attractiveness can promote increased adiposity
 - Socioeconomic factors are powerful modulators
 - Unhealthy food is often less expensive
 - Healthy food deserts are present in many areas around the world (especially urban and rural areas)
 - Breastfeeding, which is associated with lower BMI throughout the infant's life, is considered undesirable by many

The Physiology of Hunger. Lee CD, Ingelfinger JR, eds. *The New England journal of medicine*. 2025;392(4):372-381. doi:10.1056/NEJMra2402679



Physiology of Hunger – Microbiota-Driven Hunger

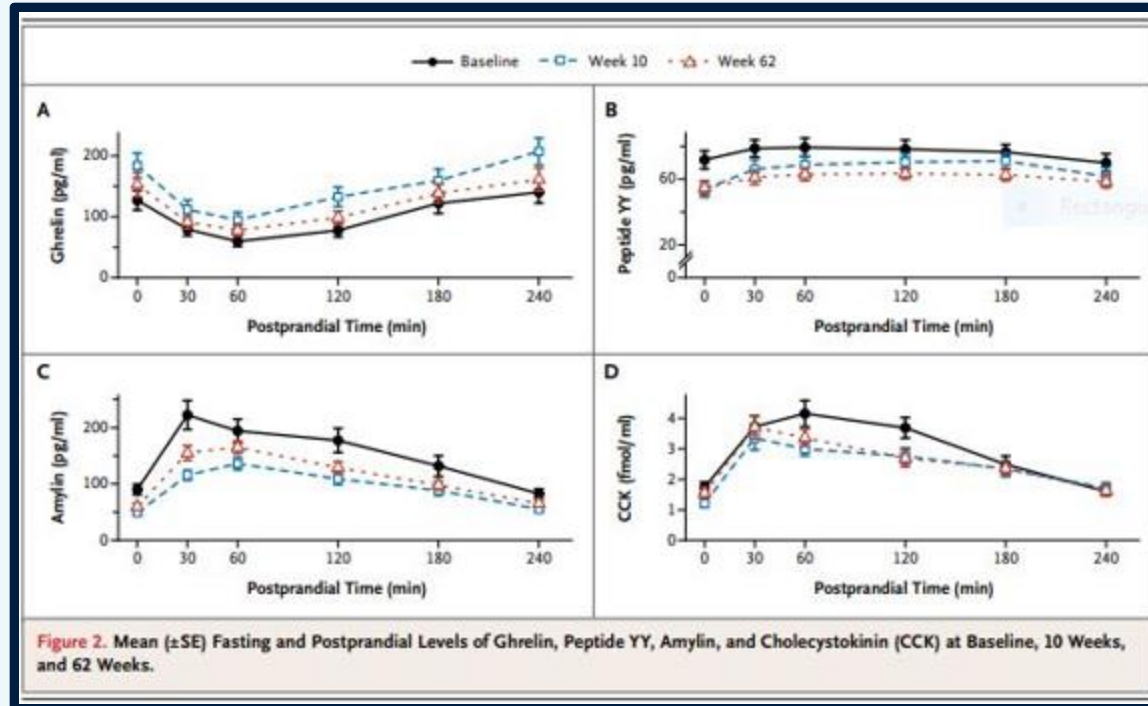
- Key Points

- Most of these data are observational involving animal models and small sample sizes - currently of limited utility
- Gut microbes capable of influencing the release of hunger and satiety hormones have been identified
 - Decreased microbiota diversity has been associated with higher serum leptin concentrations and increased insulin resistance
 - Compounds released from gut microbiota inhibit hunger through synthesis of GLP-1 and PYY and inhibition of ghrelin production
- Some gut microbes have been shown to produce peptidomimetic proteins (mimicry of endogenous hormones)
 - Some *E. coli* have been found to produce caseinolytic peptidase B (ClpB), a biomimic of α -melanocyte-stimulating hormone (α -MSH), which has been shown to increase serum concentrations of GLP-1 and PYY (satiety hormones)

The Physiology of Hunger. Lee CD, Ingelfinger JR, eds. *The New England journal of medicine*. 2025;392(4):372-381. doi:10.1056/NEJMra2402679



Metabolic Adaptations to Obesity – Energy Balance is Internally Biased



Research suggests humans maintain a fat mass set point.⁴

- The body maintains a “defended fat mass”
- There are persistent metabolic adaptations to obesity (after weight loss)
- We may help patients to lower their fat mass set point and hence, achieve weight loss
- We can modify patients’ hedonic and microbiota-related factors with lifestyle modifications, pharmacotherapy, and bariatric surgery

Long-Term Persistence of Hormonal Adaptations to Weight Loss. Sumithran, et al. *The New England Journal of Medicine*. 2011;365: 1597-1604.

Evaluation of the Patient with Overweight or Obesity - Overview

- Take a good history
- Perform a good physical exam
- Evaluate available objective data and order testing, if indicated
- Make your assessment regarding contributors, severity and complications of overweight or obesity
- Generate a treatment plan



Evaluation of the Patient with Overweight or Obesity – Take a good history

- History of present illness
 - Weight History: especially age of onset - there are monogenic, syndromic or hypothalamic forms of obesity
 - “Stress” assessment: survey for active mental health disorders (depression, etc.) and other psychologic stress
 - Sleep assessment: screen for obstructive sleep apnea and other causes of poor sleep
 - Screen for eating disorders (Binge-Eating Disorder, Bulimia Nervosa, etc.)
 - Contraception or pregnancy status: weight loss generally not desired in pregnancy; AOMs considered unsafe
 - Physical activity: relevant, but the least important



Evaluation of the Patient with Overweight or Obesity – Take a good history

- History of present illness (previous slide)
- Past medical history: pay special attention to the presence of weight-related complications
- Medications: many medications are thought to increase appetite and promote weight gain
- Allergies: presence of allergies or side effects to AOMs (anti-obesity medications) or foods
- Social history: alcohol (calories, sleep), cannabis (orexigenic), urban or rural living (food deserts)
- Surgical history: bariatric surgery or other intra-abdominal surgeries



Evaluation of the Patient with Overweight or Obesity – Perform a good exam

- General: assess muscularity and for symptoms of complications
- HEENT: evaluate for anatomy predisposing to obstructive sleep apnea, acanthosis, etc.
- Lungs: evaluate for signs of restrictive lung disease, obesity hypoventilation syndrome, etc.
- Heart: evaluate for sequelae of hypertension, arrhythmia, etc.
- Abdomen: evaluate for central adiposity, hepatomegaly, etc.
- Skin: evaluate for hirsutism, dehydration, etc.
- Extremities: evaluate for signs of lipedema, lymphedema, peripheral vascular disease, etc.



Evaluation of the Patient with Overweight or Obesity – Objective Data

- Vital signs:
 - Height
 - Weight
 - Heart rate
 - Blood pressure - make sure you use the right cuff size
 - Optional: waist circumference, body composition, temperature, pulse oximetry
- Markers of metabolic health:
 - Blood glucose +/- A1c
 - Lipid panel
 - Metabolic panel (especially kidney function parameters and liver enzymes)
 - Thyroid function testing
 - Sleep study reports
 - Liver imaging (looking for steatosis)
 - Genetic testing if there is suspicion for monogenetic or syndromic causes of overweight or obesity



Evaluation of the Patient with Overweight or Obesity – Assessment

- One-liner assessment typically contains this information:
 - Severity of illness (overweight, class 1 obesity, class 2 obesity, or class 3 obesity)
 - Presence of complications
 - Response to treatment(s)
- Example: 34 year-old male presents with class 3 obesity complicated by moderate restrictive lung disease, obesity hypoventilation syndrome with severe pulmonary hypertension, early coronary artery disease, history of acute myocardial infarction and type 2 diabetes. He has lost 4% total body weight with lifestyle modifications and tirzepatide. He is experiencing intolerable side effects to tirzepatide.
- Example: 48 year-old female presents with class 2 obesity complicated by primary hypertension and mild obstructive sleep apnea. She has lost 13% total body weight with lifestyle modifications and semaglutide which she is tolerating well.



Evaluation of the Patient with Overweight or Obesity – Generate a Plan

- Treatment of overweight and obesity requires an interdisciplinary, multimodal approach:
 - Lifestyle modifications - all members of the team
 - Medical nutrition therapy - Registered Dietitian
 - Behavioral therapy - LCSW, PsyD, clergy, etc.
 - Anti-obesity medication (AOM) - physician or advanced practice clinician (NP or PA)
 - Bariatric surgery - physician or advanced practice clinician



Treatment of Obesity – Lifestyle Modifications

- Comprehensive Lifestyle Modifications
 - Sleep optimization
 - Stress reduction and management
 - Nutrition optimization
 - Physical activity optimization

Treatment of Obesity - Lifestyle Modifications

- Medical nutrition therapy can be very helpful (in some cases, it is required)
 - Typically given by a Registered Dietitian (RD), physician, NP, PA or Registered Diabetes Education (RDE, often RNs)
 - Anyone can call themselves a “nutritionist”
 - RDs take board exams and must maintain licenses to practice
- Unfortunately, not all patients have access to a Registered Dietitian
 - Some health insurance plans do not cover visits with an RD unless a patient has certain diagnoses
 - Many patients live in areas where there are no Registered Dietitians within a reasonable distance
 - Many patients do not have access to reliable transportation
- Nutrition science is a field plagued by mixed results and industrial bias
- You should seek sources of reliable online information (diabetes.org, and others)
 - Please maintain a healthy skepticism about information obtained from social media platforms
- Many electronic medical records have decent databases of patient handouts with nutrition information



Treatment of Obesity - Lifestyle Modifications

- Behavioral therapy can be very helpful (in some cases, it is absolutely necessary for weight loss)
 - Typically provided by an LCSW with training in therapy, psychologist, clergy or other mental health professional
- Unfortunately, not all patients have access to behavioral therapy
 - Some health insurance plans do not cover visits with a therapist
 - Some patients live in areas where there are no behavioral therapists within a reasonable distance
- Many behavioral health providers offer their services by telemedicine



Anti-Obesity Medications (AOMs): General Approach

- Pharmacotherapy should always be used in conjunction with comprehensive lifestyle modifications.
- Pharmacotherapy is indicated in non-pregnant adults with BMI ≥ 27 kg/m² with life-threatening comorbidity (hypertension, diabetes, obstructive sleep apnea, heart disease, etc.) and all patients with BMI ≥ 30 kg/m².
- Generally, pharmacotherapy is considered if there is <5% total body weight loss (TBWL) after 3 months of lifestyle modifications alone.
- Generally, pharmacotherapy is changed, discontinued or up-titrated if there is <5% TBWL after 3 months of use.



Anti-Obesity Medications: General Approach

- We use a dose-escalation strategy: initiate at a low dose and titrate to the minimum dose needed to achieve the desired outcome(s).
- Although it is off-label for some of our AOMs, we typically use these medications in the long-term (think of them like antihypertensives or antihyperglycemics).
- No AOM is considered safe in pregnancy



Anti-obesity Medications: Lipase Inhibitor

- **Orlistat**
 - **Mechanism of Action:** lipase inhibitor resulting in decreased capacity to breakdown and absorb lipids within the GI tract
 - **Contraindications:** cholestasis, chronic malabsorption syndrome, hypersensitivity, pregnancy
 - **Common side effects:** abdominal discomfort, abdominal pain, fecal urgency, anal leakage, steatorrhea
 - *Note: Alli is a lower-dose over-the-counter formulation, Xenical is a higher-dose prescription formulation*



Anti-obesity Medications: Stimulants (sympathomimetic amines)

- **Phentermine, Diethylpropion**

- **Mechanism of Action:** stimulate norepinephrine release resulting in appetite suppression (through an unknown mechanism)
- **Contraindications:** cardiovascular disease, glaucoma, hyperthyroidism, uncontrolled hypertension, hypersensitivity, pregnancy
- **Common side effects:** agitation, anxiety, constipation, insomnia, tachycardia, xerostomia
- **Rare, but serious, side effects:** arrhythmia, acute glaucoma, acute coronary syndrome
- *Note: studies regarding side effects are conflicting; seems to have low abuse potential*



Anti-obesity Medications: Combined Substance Use Medications

- **Naltrexone-bupropion ER**

- **Mechanism of Action:** opioid antagonist, dopamine/norepinephrine reuptake inhibitor resulting in appetite suppression (through unknown mechanisms)
- **Contraindications:** concomitant opioid use (naltrexone may precipitate withdrawal), seizure disorder (bupropion lowers seizure threshold), uncontrolled hypertension, hypersensitivity, pregnancy
- **Common side effects:** constipation, nausea, vomiting, dizziness, insomnia, headache
- *Note: naltrexone is indicated separately for alcohol and opioid use disorders; bupropion is indicated separately for smoking cessation, major depression, and seasonal affective disorder*



Anti-obesity Medications: Combined Stimulant-Antiepileptic

- **Phentermine-topiramate ER**

- **Mechanism of Action:** stimulate norepinephrine release and modulates CNS GABA receptors resulting in appetite suppression (through unknown mechanisms)
- **Contraindications:** cardiovascular disease, glaucoma, hyperthyroidism, uncontrolled HTN, MAOI use within 14 days, hypersensitivity, pregnancy
- **Common side effects:** agitation, insomnia, xerostomia, constipation, altered taste, paresthesia, dizziness
- *Note: topiramate is also indicated as an antiepileptic, mood stabilizer, and for migraine prevention*



Anti-obesity Medications: GLP-1 Agonists

- **Liraglutide, Semaglutide, Orforglipron**
 - **Mechanisms of Action:** GLP-1 receptor agonists indirectly stimulate POMC neurons in the arcuate nucleus of the hypothalamus promoting satiety, inhibit the limbic (reward) system, and directly stimulate GLP1Rs throughout GI tract to slow gastric emptying (among other actions)
 - Refer to DM2 pharmacology lecture for contraindications and side effects
 - *Note: Saxenda is a once-daily injection*
 - *Note: Wegovy is a once-weekly injection and is also FDA-approved for cardiovascular risk reduction in people with a history of myocardial infarction, history of stroke or diagnosis of peripheral artery disease*
 - *Wegovy is also FDA-approved for treatment of Metabolic dysfunction-Associated Steatohepatitis (MASH) with moderate or advanced fibrosis*



Anti-obesity Medications: GLP-1/GIP Agonist

- **Tirzepatide**

- **Mechanism of Action:** GLP-1/GIP receptor agonists indirectly stimulate POMC neurons in the arcuate nucleus of the hypothalamus promoting satiety, inhibit the limbic (reward) system, and directly stimulate GLP1Rs throughout GI tract to slow gastric emptying (among other actions)
- Refer to DM2 pharmacology lecture for contraindications and side effects
- *Note: Zepbound is a once-weekly injection also FDA-approved for the treatment of moderate or severe obstructive sleep apnea*



Anti-obesity Medications: Melanocortin-4 Receptor Agonist

- **Setmelanotide**

- Indicated for severe obesity caused by certain genetic disorders in patients ≥ 2 years old
 - Bardet-Biedl Syndrome
 - POMC Receptor Mutation
 - PCSK1 Mutation
 - Leptin Receptor Mutation
- **Mechanism of Action:** Melanocortin-4 receptor (MC4R) agonist resulting in decreased feeding behavior and increased energy expenditure
- **Contraindications:** history of hypersensitivity reaction
- **Most common side effects:** hyperpigmentation, increased libido, hypersensitivity
- *Note: Imcivree is a once-daily injection*



Anti-obesity Medications: Efficacy Summary

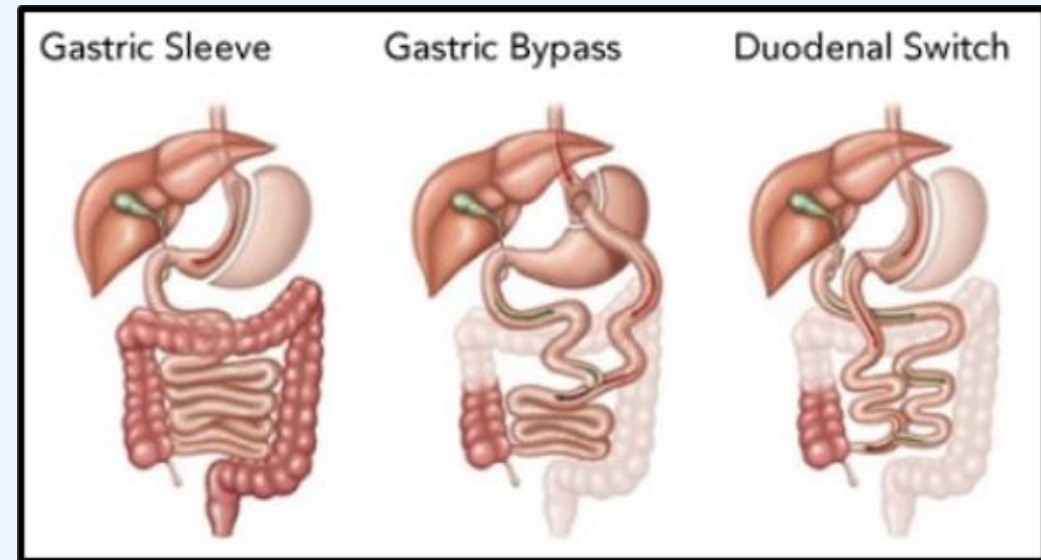
Anti-obesity Medication	Efficacy (%TBWL)	Relative Cost
Orlistat	3	\$
Naltrexone-Bupropion	4	\$\$
Phentermine	7	\$
Liraglutide	8	\$\$
Phentermine-Topiramate	8	\$\$
Orforglipron	12	\$\$\$
Setmelanotide	13	\$\$\$\$\$
Semaglutide oral	14	\$\$\$
Semaglutide injection	16	\$\$\$
Tirzepatide	20	\$\$\$

Effectiveness and safety of drugs for obesity. Lewis, et al. *The British Medical Journal*. 2024;384:e072686.



Treatment of Obesity – Bariatric Surgery

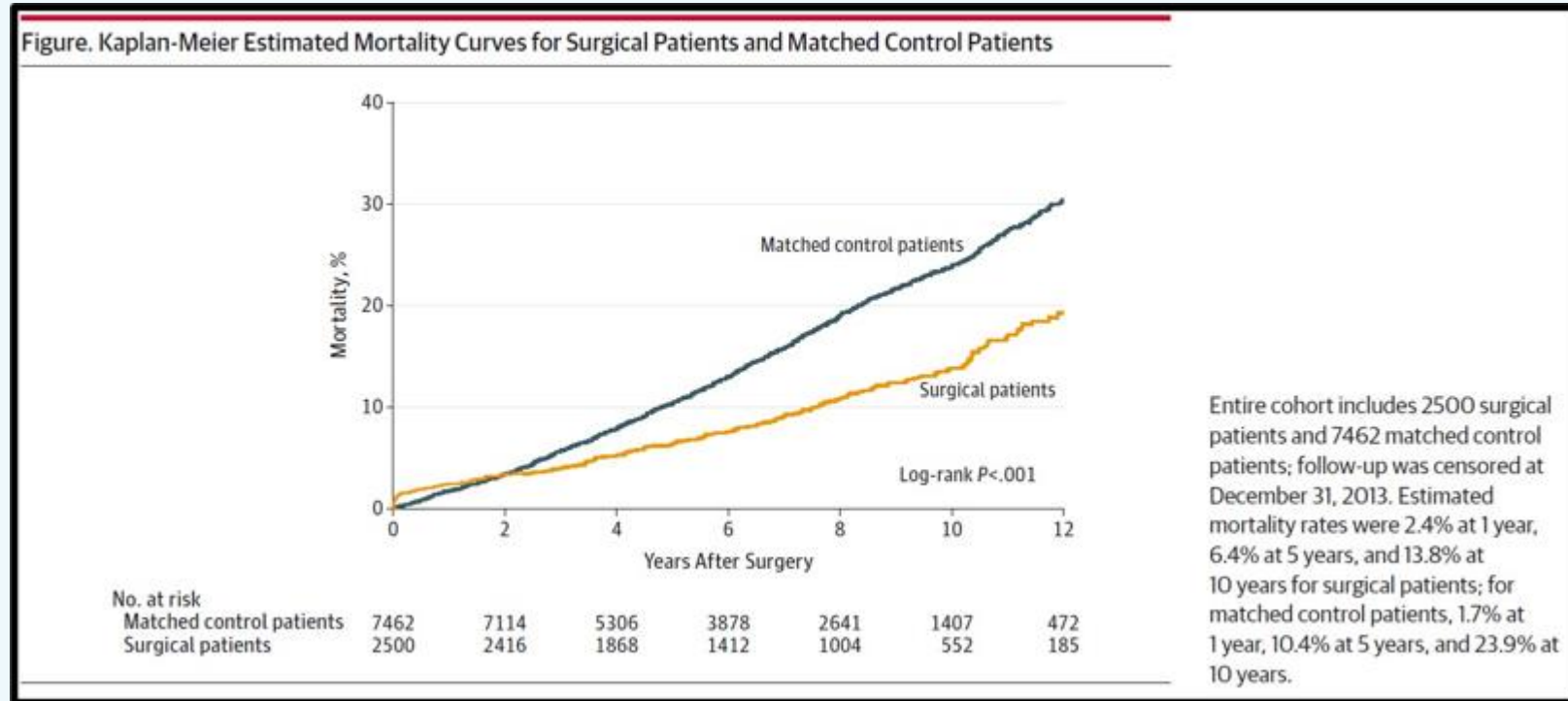
- Bariatric Surgery is an important option and, for many patients, the only way they will experience, and maintain, enough weight loss to achieve improved health
- Current criteria for bariatric surgery (according to the ASMBS and IFSO):
 - All patients BMI ≥ 30 kg/m² with metabolic complications of obesity (diabetes, hyperlipidemia, hypertension, etc.)
 - All patients BMI ≥ 35 kg/m²
 - *Note: most insurance companies do not use these as coverage criteria*
- Lifelong vitamin supplementation is required



2022 American Society for Metabolic and Bariatric Surgery (ASMBS) and International Federation for the Surgery of Obesity and Metabolic Disorders (IFSO): Indications for Metabolic and Bariatric Surgery
Eisenberg, Dan et al. Surgery for Obesity and Related Diseases, Volume 18, Issue 12, 1345 - 1356

Treatment of Obesity – Bariatric Surgery

- Bariatric surgery saves lives



Arterburn, et al. (2014) Association Between Bariatric Surgery and Long-term Survival. JAMA (313) 62-70

Summary

- Obesity is an highly-prevalent disease resulting from aberrations in the complex physiology of hunger and energy balance.
- Our physiology evolved to favor overconsumption of energy and the modern environment promotes the same.
- Researchers have identified complex orexigenic (weight gain-favoring) and anorexic biochemical pathways involving, primarily, the gut and brain (most notably the hypothalamus).
 - There are rare monogenetic and syndromic causes of obesity resulting in aberrations in these pathways.



Summary

- Humans exhibit evidence of a “defended fat mass” and experience metabolic adaptations to obesity.
- The primary physiologic mechanisms of hunger are the homeostatic, hedonic, and microbiota-driven mechanisms.
- In modern settings, the hedonic hunger mechanism predominates. Hedonic hunger can be triggered by many different stimuli.



Summary

- BMI is widely used to identify those with excess adiposity, but it is a flawed metric.
 - BMI 25-29.9kg/m² is considered overweight, BMI ≥30kg/m² is considered obesity
- Treatment of obesity requires a patient-centered, team-based, interdisciplinary, multimodal approach.
- Treatment options for obesity include comprehensive lifestyle modifications, anti-obesity medications, and bariatric surgery.
- Comprehensive lifestyle modifications include sleep optimization, stress management, diet modification, obesogenic medication avoidance, and physical activity optimization.



Summary

- Naltrexone-bupropion, phentermine, phentermine-topiramate, and orlistat are FDA-approved oral AOMs of lower effectiveness and cost.
- Liraglutide, orforglipron and semaglutide are GLP-1 receptor agonists and FDA-approved AOMs of moderate effectiveness and cost.
- Tirzepatide is a GLP-1/GIP receptor agonist and FDA-approved AOM of higher effectiveness and cost.
- Bariatric surgery is a safe, and effective, treatment for obesity that has been shown to promote survival.
 - Bariatric surgery is indicated for patients of BMI ≥ 30 kg/m² with metabolic complications of obesity (diabetes, hyperlipidemia, hypertension, etc.) and all patients of BMI ≥ 35 kg/m².



Questions?

Thank you for your attention.

Please feel free to email me at nathan.lent@jefferson.edu if you have any questions.